# KUKU — AI GENERALIST ASSIGNMENT

**Problem 1: Find the Promos (Ads)** | Candidate Solution & Technical Architecture

## 1. Problem Choice, Strategic Context & Framing

**Why Problem 1?** Microdramas are a high-velocity, acquisition-driven format. The bottleneck for organic and paid growth on TikTok, Instagram Reels, and YouTube Shorts is manual video clipping: human editors spend hours per season scrubbing timelines to find 20–60s hooks. Automating this directly impacts top-of-funnel customer acquisition cost (CAC) and throughput.

**Key Assumptions Made:**

• *Strict Duration Enforcement:* Clips must respect platform drop-off curves (optimal 25–45s; hard range 20–60s).

• *Continuous vs Multi-Segment:* While cross-episode montages exist, 90%+ of viral microdrama ads are single continuous dramatic beats that preserve raw emotional momentum without jarring visual cuts.

• *Zero-Lore Comprehension:* A top-of-funnel ad must be understandable to a cold scroller within 3 seconds without knowing previous plotlines.

## 2. Technical Architecture: Two-Stage Hybrid Engine

To balance operational cost, latency, and nuanced creative reasoning at real-world scale, we rejected a naive 'throw all 50 scripts at a closed LLM' approach. Instead, we architected a **Two-Stage Hybrid Pipeline**:

| Component | Deterministic Code / Simple Rules | Generative AI / LLM Layer |
|---|---|---|
| **Responsibilities** | • Screenplay parsing & regex extraction of millisecond timestamps.<br>• Sliding window generation (20s–60s constraints).<br>• Anchor detection: Dialogue starts, high-intensity tones (Cold, Furious, Desperate), slap/confrontation keywords.<br>• Mathematical scoring of 0–3s hook & final cliffhanger. | • Semantic tension assessment & context check.<br>• Generates punchy social-media rationale ('Why it works').<br>• Clickbait title generation and marketing angle identification.<br>• Swappable API adapter (Gemini 2.5 Flash / OpenAI / Claude). |
| **Cost & Scale** | **$0.00 base cost**; runs in <2.0s across 50 episodes on standard CPU. | Only evaluates top filtered candidate windows (~3–4/ep). Cost: **~$0.008 per show**. |

## 3. Scoring Rubric & Window Selection Logic

Candidate clips are evaluated on a composite 100-point scale across three weighted dimensions:

• **Hook Score (40%):** Evaluates the first 3.0 seconds. Looks for high-stakes disruption verbs (*'marry', 'turn around', 'betray', 'enough'*), confrontational tones (*Venomous, Disgusted, Screaming*), or physical shock actions (*slap, tear, doors burst*).

• **Cliffhanger Score (35%):** Evaluates the final 3.0–5.0 seconds. Looks for unresolved interrogatives (*'?'*), interrupted statements (*'...'*), acoustic swells (*'ominous cliffhanger peak'*), or visual freeze/cut-to-black.

• **Standalone Coherence (25%):** Requires >= 2 distinct speakers in conflict and 4–10 dialogue turns. Penalizes monologue exposition.

## 4. Strategic Campaign Rollout: Which Promos to Post First & Why?

Ranking microdrama promos is not just about raw emotional score; it requires a **funnel-aligned publishing strategy**:

| Tier & Priority | Target Episodes | Strategic Rationale & Placement |
|---|---|---|
| **Tier 1: Post First** (Cold User Acquisition) | **Episodes 1 – 3** (Rank #1: Ep 3, 00:25–00:56 Rank #2: Ep 1, 01:15–01:38) | **Zero-Lore Universality:** The wedding betrayal, sister stealing the groom, and mysterious hero's intervention require zero prior knowledge. Highest conversion from cold scroller to initial app install. |
| **Tier 2: Post Second** (Viral Escalation & Shares) | **Episodes 4 – 15** (Identity reveals & public confrontation) | **Power Reversal & Catharsis:** Juliana standing her ground and the Duke's true status emerging. Drives high comment velocity and algorithmic engagement on TikTok/Instagram. |
| **Tier 3: Post Third** (Retargeting & Churn Prevention) | **Episodes 16 – 50** (High-cliffhanger mid-season) | **Binge Triggers:** Life-or-death and high-stakes ultimatums targeted at warm audiences who installed the app but stalled at Episode 5. |

## 5. Honest Quality Evaluation & Failure Modes ('Where does the tool pick badly?')

We ran an automated quality audit (`src/quality_checker.py`) across all 100 extracted promos from the 50 episodes. This surfaced **4 specific failure modes**:

• **1. Premature Cut Buffer (58% of clips):** The script marks dialogue end at e.g. 00:32.40. Cutting exactly at that timestamp clips the actor's stunned facial reaction or the sound sting. *Mitigation:* Our parser adds an automated +0.75s visual reaction buffer.
• **2. Off-Target Secondary Characters (32% of clips):** Heated shouting matches between maids or secondary aristocrats score high on acoustic/lexical aggression, but fail as ads because scrollers lack empathy for peripheral characters. *Mitigation:* Implemented protagonist-weighting bonus.
• **3. Lore & Exposition Overload:** Scenes debating estate ledgers or aristocratic lineage score high on dialogue balance but lack immediate visual/emotional peril. *Mitigation:* Average-words-per-turn threshold flag (>25 words/turn flagged for review).
• **4. Visual-Only Ambiguity:** Pure action moments (sword draws, silent glares) lack dialogue cues, leading text heuristics to underestimate them. *Mitigation:* Multi-modal expansion roadmap.

## 6. Production Scaling, Cost Analysis & Deliverables

• **Cost at Scale:** Processing 20 new shows/week (1,000 episodes) using our hybrid approach costs **under $0.16/week** in API calls ($0 on base tier) and takes under 45 seconds of compute.
• **Human-in-the-Loop Workflow:** The tool acts as an AI Copilot for video editors—filtering 100+ minutes of footage down to the top 3 high-confidence candidate clips with pre-calculated in/out timecodes, reducing promo turnaround from 4 hours to 10 minutes.
• **Deliverables Provided:** (1) *Streamlit Web Dashboard* with live interview runner; (2) *promos_master.json* & *promos_ranked.csv* covering all 50 episodes; (3) *GitHub Repository* with 1-command reproducibility.